



Traditional telecom antenna

not wasted in antenna reflection.

There is a long list of antenna designs with their suitability, advantages and limitations. There are many antenna types and many ways of categorising them. Antenna types can be used to differentiate antennas for radios, televisions and radar systems. Because antennas can be built for transmission of different frequencies, another way to categorise antenna types is by their frequency. For radio antennas, it's important to know whether these are built for, say, frequency modulation (FM) broadcasting at 88-108 MHz or amplitude modulation (AM) broadcasting at 535-1605 kHz. For television antennas, one distinguishes between ultra-high frequency (UHF) antennas and very-high frequency (VHF) antennas, or antennas that pick up both.

Stores that sell antennas categorise various types on terms of customers' needs. The range of antennas can be categorised as short, medium or long. For customers buying a television antenna, the decision is dependent on how close they are to the transmitting towers that they wish to pick up a signal from. If the range is well-matched to the distance, it will help avoid the antenna picking up unwanted signals.

Location is another way of looking

at antenna type. Antennas can be made for indoor, outdoor or attic installation. Indoor antennas are easy to install but usually do not have the elevation to provide the best signal, particularly for customers who are far from the transmission. Outdoor antennas were primarily made for rooftops, but now more are being designed to mount on the side of a house, on a pole or deck. The attic

can be a useful installation point for those who do not want their antenna inside or outside for aesthetic or other reasons. Another set of antenna types is differentiated by style. Style can mean the antenna's appearance in terms of design. It can also address whether the antenna is directional and gathers signals from a central location or whether it is multi-directional—seeking signals from towers transmitting from different locations. The latest version of antenna, i.e., plasma antenna employs ionised gas enclosed in a tube (or other enclosure) as the conducting element of the antenna.

Plasma antennas

The different states of matter generally found on earth are solid, liquid and gas. Sir William Crookes, an English physicist, identified a fourth state of matter, now called plasma, in 1879. Plasma is by far the most common form of matter. Plasma in the stars and in the tenuous space between them makes up over 99 per cent of the visible universe and perhaps most of what is not visible.

Important to antenna technology, plasmas are conductive assemblies of charged and neutral particles and fields that exhibit collective effects.

Plasmas carry electrical currents and generate magnetic fields.

A plasma antenna is a type of antenna in which the metal-conducting elements of a conventional antenna are replaced by plasma. These are radio frequency antennas that employ plasma as the guiding medium for electromagnetic radiation. The plasma antennas are essentially a cluster of thousands of diodes on a silicon chip that produces a tiny cloud of electrons when charged. These tiny, dense clouds can reflect high-frequency waves like mirrors, focusing the beams by selectively activating particular diodes. The 'beam-forming' capability could allow ultra-fast transmission of high data loads—like those needed to seamlessly stream a TV show to an untethered tablet—creating an attractive option for the next generation of supercharged wireless transmitters.

Many types of plasma antennas can be constructed, including dipole, loop and reflector antennas. Plasma antennas are interpreted as various devices in which plasma with electric conductivity serves as an emitting element. In gas plasma antenna the concept is to use plasma discharge tubes as the antenna elements. When the tubes are energised, these turn into conductors, and can transmit and receive radio signals. When de-energised, these revert to non-conducting elements and do not reflect probing radio signals.

The fact that the emitting element is formed over the interval needed for the emission of an electromagnetic pulse is an important advantage of plasma antennas. In the passive state (in the absence of plasma in the discharge tube), such a device does not exhibit electric conductivity.

A plasma stream flowing from a jet into the ambient space, the plasma trace of a body moving at an ultrasonic velocity in the atmosphere, and alternative plasma objects have been studied as possible antenna elements. Solid-state plasma antenna uses beam-forming technology and the same manufacturing process that is currently